

Zulfiya Usmonova

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Research Interests: AI Evaluation & Benchmarking — LLM Robustness — Human-AI Interaction — Multi-Agent Systems

EDUCATION

Technical University of Munich (TUM)

Munich, Germany

MSc in AI in Society, GPA: 1.2 (1st semester)

Oct. 2025 – March. 2027

- Coursework: Human-AI Interaction, Foundations of AI and Data Science, AI in Metaverse, Introduction to Deep Learning, Path to AGI, Psychology of Learning and Instructional Design of AI-based Systems .

New Uzbekistan University

Tashkent, Uzbekistan

BSc in Software Engineering, GPA: 4.1/4.5 (Rank: 6/115)

Oct. 2021 – Jun. 2025

- Honor of University Award (2023) and Top Student Award (2022).
- Engaged in Topological Data Analysis seminars and guest lectures from TUM, Cornell, and MIT faculty.

Comenius University in Bratislava

Bratislava, Slovakia

Exchange Semester (NSP Scholarship Recipient)

Feb. 2025 – Jun. 2025

- Actively volunteered in conferences and seminars hosted by AI industry professionals (e.g., Google DeepMind founders, Student Conferences).
- Coursework: Neural Networks, Statistical Methods in AI, Combinatorics & Graph Theory, Discrete Math 2, Biostatistics

PUBLICATIONS & ACADEMIC WRITING

Submitted manuscript, EMNLP 2026 (Supervised by Prof. E. Kasneci)

Nov. 2025 – May 2026

When Meaning Decays: Semantic Drift in LLM Transmission Chains

TUM, Germany

- Designed and implemented a controlled 10-step transmission-chain framework to evaluate semantic drift across five open-weight LLMs, two experimental pipelines, and 400 English concepts.
- Developed a multi-metric evaluation pipeline combining exact recovery, embedding-based semantic similarity, entropy, semantic-sink analysis, predictive modeling, and a 100-participant human reference study.
- Compared two ways of passing information across repeated LLM interactions and found that meaning became less stable over time, particularly when text was repeatedly paraphrased.

Book Chapter (Supervised by Prof. M. U. Ayoobkhan)

Dec. 2023 – Apr. 2024

CRC Press, Taylor&Francis Group

Tashkent, Uzbekistan

- Cloud Computing for Smart Education, Transformative Pedagogy: Collaborative Learning with Cloud Architecture and Virtual Programming Labs.
- Abstract available: <https://bit.ly/book-chapter-link>

RESEARCH & WORK EXPERIENCE

Student Research Assistant

Nov. 2025 – Present

Chair for Human Centered Technologies for Learning

TUM, Germany

- Designing and implementing a prompted weak supervision pipeline using local large language models for automated teaching quality assessment.
- Prototyping and benchmarking local open-weight LLMs and setting up server-based experimentation workflows and lightweight model-testing pipelines for scalable and reproducible evaluation.
- Conducting a systematic literature review of personalized AI coaching agents in immersive learning environments, synthesizing findings from more than 100 studies.

TUM Social AI Club - AI Engineer

Oct. 2025 – Present

Collaboration with InitiativGruppe and UN Women Germany

Munich, Germany

- Built and deployed an LLM-based chatbot for a public-facing website, implementing safety-oriented prompting strategies and response validation for reliable user interaction.
- Developing a benchmark suite to evaluate LLM responses on women's rights and gender-equality scenarios, measuring factual accuracy, safety, stereotype resistance, framing fairness, and consistency.

AI Research / Engineering Project

Nov. 2025 – Jan. 2026

Multi-Agent LLM Backend for Metaverse Simulation

TUM, Germany

- Designed and implemented a Python Flask backend linking a Godot-based simulation with locally hosted LLMs, enabling heterogeneous multi-agent architectures with memory for autonomous decision-making in a shared virtual environment.
- Debugged and stabilized the system under concurrent requests, addressing timeouts, model latency, and API incompatibilities to ensure reliable agent execution.
- Authored comprehensive technical documentation covering system architecture, API integration, setup, deployment, and troubleshooting to support reproducibility and future development.

Bachelor's Thesis

Sep. 2024 – Apr. 2025

When AIs Change Their Minds: Testing LLM Robustness to Altered Voting Protocols

Tashkent, Uzbekistan

- Built a 100-agent simulation across 10 LLM families to test robustness under altered voting protocols.
- Designed a two-stage study: (i) individual baseline establishes per-model agree/disagree rates; (ii) network stage embeds agents in random d -regular graphs ($d = 1 - 9$) and measures vote-flip rates as peer exposure grows.
- Discovered that modest network degree ($d > 1$) can invert the majority for smaller instruction-tuned models (phi, Gemma) while larger, alignment-optimized models (GPT-4, Gemini) remain stable.
- Code and report: <https://github.com/UsmonovaZulfiya/bachelor-s-thesis.git>

SKILLS

Programming: Python, Git, Unity, SQL, R

Libraries & Frameworks: Scikit-learn, Pandas, Matplotlib, NetworkX, LangGraph, AutoGen, PyTorch

Agent Architectures: tool calling, memory (STM/LTM), multi-agent orchestration

LLM Ops: embeddings/similarity metrics, local LLM serving, OpenAI-compatible APIs, benchmarking

Backend: Flask APIs, concurrent request handling, JSON-based protocols

SELECTED COMPETITIONS AND VOLUNTEERING

Competitions & Hackathons: Huawei Seeds for the Future Global Competition (2nd place), NASA Space Apps Challenge Uzbekistan (Winner)

Volunteering: Women In Tech Uzbekistan Chapter Marketing Team Lead (2025–Present); Huawei ICT Ambassador (2024–25); TechnovationGirls Mentor and Ambassador (2022–24); Academic Mentor at Summer School (2025); Education Minister & General Secretary, NewUU Student Council (2022–24).